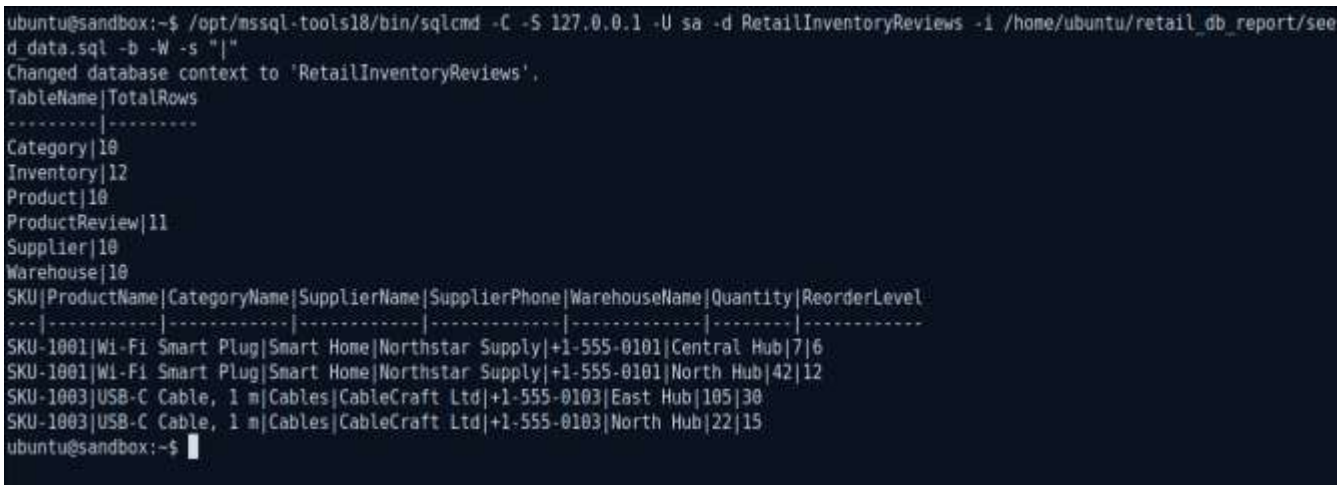
Day 5 assignment -SQL &MongoDB

1. Normalized data query

```
SELECT p.SKU, p.ProductName, c.CategoryName, s.SupplierName, s.SupplierPhone, w.WarehouseName,
i.Quantity, i.ReorderLevel
FROM dbo.Product AS p
JOIN dbo.Category AS c ON c.CategoryID = p.CategoryID
JOIN dbo.Supplier AS s ON s.SupplierID = p.SupplierID
JOIN dbo.Inventory AS i ON i.ProductID = p.ProductID
JOIN dbo.Warehouse AS w ON w.WarehouseID = i.WarehouseID
WHERE p.SKU IN (N'SKU-1001', N'SKU-1003')
ORDER BY p.SKU, w.WarehouseName;
```

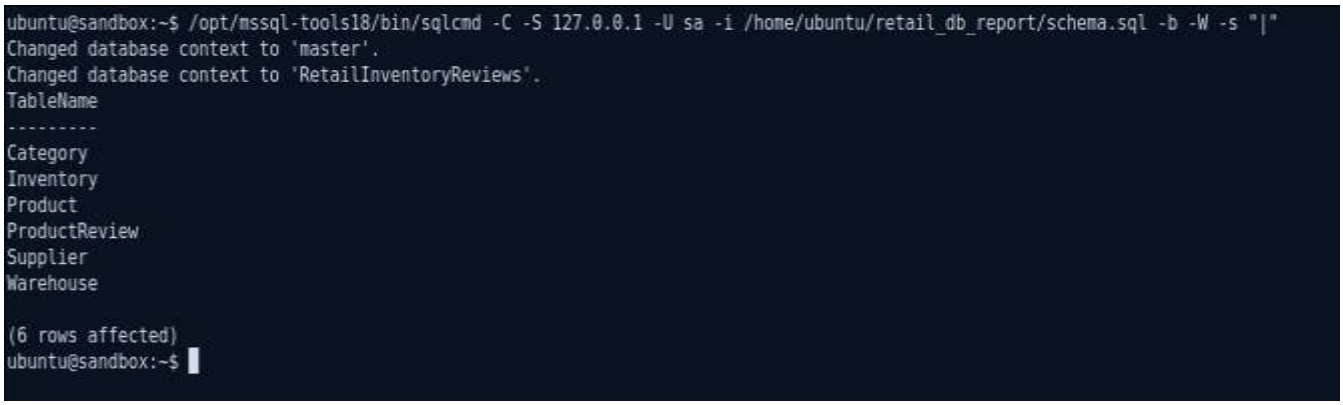
Output screenshot



2. SQL Server schema check

```
SELECT t.name AS TableName
FROM sys.tables AS t
WHERE t.is_ms_shipped = 0
ORDER BY t.name;
```

Output screenshot



3. Seed row counts and failing inserts

```
SELECT N'Category' AS TableName, COUNT_BIG(*) AS TotalRows FROM dbo.Category
UNION ALL SELECT N'Supplier', COUNT_BIG(*) FROM dbo.Supplier
UNION ALL SELECT N'Warehouse', COUNT_BIG(*) FROM dbo.Warehouse
UNION ALL SELECT N'Product', COUNT_BIG(*) FROM dbo.Product
UNION ALL SELECT N'Inventory', COUNT_BIG(*) FROM dbo.Inventory
UNION ALL SELECT N'ProductReview', COUNT_BIG(*) FROM dbo.ProductReview
ORDER BY TableName;

INSERT dbo.Category (CategoryName) VALUES (N'Smart Home'); -- duplicate unique
INSERT dbo.Product (SKU, ProductName, CategoryID, SupplierID) VALUES (N'SKU-BAD-PARENT', N'Missing
category test', 999, 1);
INSERT dbo.Inventory (ProductID, WarehouseID, Quantity, ReorderLevel) VALUES (1, 10, -5, 3);
INSERT dbo.Supplier (SupplierName, SupplierPhone) VALUES (NULL, N'+1-555-0199');
```

Output screenshot

```
ubuntu@sandbox:~$ /opt/mssql-tools18/bin/sqlcmd -C -S 127.0.0.1 -U sa -d RetailInventoryReviews -i /home/ubuntu/retail_db_report/see
d_data.sql -b -W -s "|"
Changed database context to 'RetailInventoryReviews'.
TableName|TotalRows
---------|---------
Category|10
Inventory|12
Product|10
ProductReview|11
Supplier|10
Warehouse|10
SKU|ProductName|CategoryName|SupplierName|SupplierPhone|WarehouseName|Quantity|ReorderLevel
---|-----------|------------|------------|------------|-------------|--------|------------
SKU-1001|Wi-Fi Smart Plug|Smart Home|Northstar Supply|+1-555-0101|Central Hub|7|6
SKU-1001|Wi-Fi Smart Plug|Smart Home|Northstar Supply|+1-555-0101|North Hub|42|12
SKU-1003|USB-C Cable, 1 m|Cables|CableCraft Ltd|+1-555-0103|East Hub|105|30
SKU-1003|USB-C Cable, 1 m|Cables|CableCraft Ltd|+1-555-0103|North Hub|22|15
ubuntu@sandbox:~$
```

Output screenshot

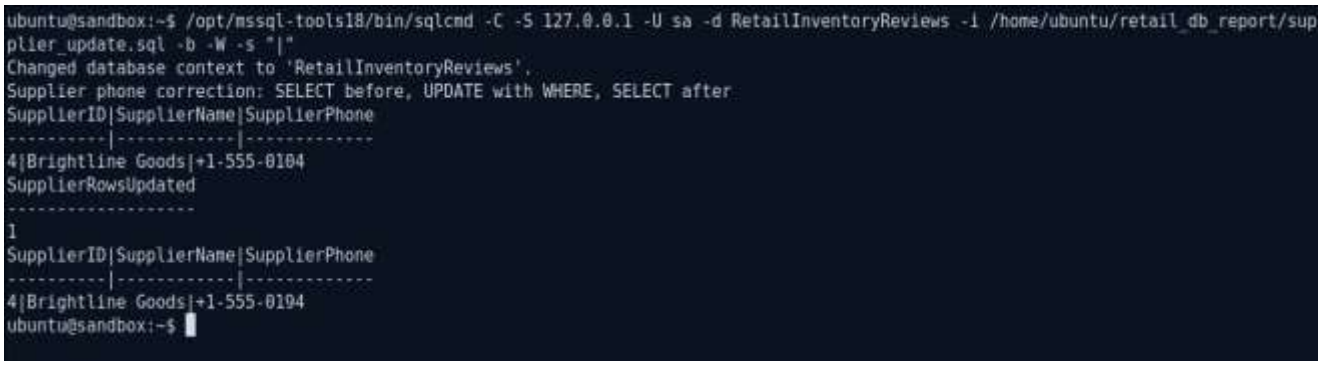
```
ubuntu@sandbox:~$ /opt/mssql-tools18/bin/sqlcmd -C -S 127.0.0.1 -U sa -d RetailInventoryReviews -i /home/ubuntu/retail_db_report/con
straint_failures.sql -b -W -s "|"
Changed database context to 'RetailInventoryReviews'.
Four deliberately failing INSERT tests
Test|ErrorNumber|ActualServerMessage
----|-----------|------------------
Duplicate unique value|2627|Violation of UNIQUE KEY constraint 'UQ_Category_CategoryName'. Cannot insert duplicate key in object 'db
o.Category'. The duplicate key value is (Smart Home).
Test|ErrorNumber|ActualServerMessage
----|-----------|------------------
Missing foreign-key parent|547|The INSERT statement conflicted with the FOREIGN KEY constraint "FK_Product_Category". The conflict o
ccurred in database "RetailInventoryReviews", table "dbo.Category", column 'CategoryID'.
Test|ErrorNumber|ActualServerMessage
----|-----------|------------------
Negative quantity|547|The INSERT statement conflicted with the CHECK constraint "CK_Inventory_Quantity_Nonnegative". The conflict oc
curred in database "RetailInventoryReviews", table "dbo.Inventory", column 'Quantity'.
Test|ErrorNumber|ActualServerMessage
----|-----------|------------------
NULL mandatory value|515|Cannot insert the value NULL into column 'SupplierName', table 'RetailInventoryReviews.dbo.Supplier'; colum
n does not allow nulls. INSERT fails.
ubuntu@sandbox:~$
```

4. Supplier UPDATE and discontinued SKU DELETE

```
UPDATE dbo.Supplier
SET SupplierPhone = N'+1-555-0194'
WHERE SupplierID = 4 AND SupplierPhone = N'+1-555-0104';

DELETE FROM dbo.Product
WHERE IsDiscontinued = 1;
```

Output screenshot



Output screenshot



5. MongoDB validator and CRUD

```
const retailDb = db.getSiblingDB("RetailReviewsMongo");
retailDb.createCollection("reviews", {validator: {$jsonSchema: {
  bsonType: "object", required: ["sku", "reviewText"], additionalProperties: false,
  properties: {sku: {bsonType: "string"}, reviewText: {bsonType: "string"}}
}}, validationLevel: "strict", validationAction: "error"});
const reviews = retailDb.getCollection("reviews");
reviews.insertMany(sampleReviews);
reviews.find({sku: "SKU-1001"}).sort({reviewDate: -1}).toArray();
reviews.updateOne({sku: "SKU-1001", reviewer: "alex_r"}, {$set: {reviewText: "Setup took two minutes;
app pairing was straightforward.", lastUpdated: new Date()}});
reviews.deleteOne({sku: "SKU-1010", reviewer: "robin_d"});
```

Output screenshot

